

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=milk

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.012483062457831949	0.01	4.729732285866721e-13	0.1	1.068911264269411e-21	0.1	I(1)	1.0	milk	silpo_novus	promo_controlled	nan	nan
prozorro	0.0014111002754477221	0.01	3.8189780115701814e-13	0.1	4.4125656034645e-11	0.1	I(1)	0.0	milk	silpo_novus	promo_controlled	nan	nan
retail	0.9135252706786217	0.08526130462523543	0.1118185172729621	0.1	0.0034682897791077313	0.1	I(2)	0.0	milk	silpo_novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo_novus	promo_controlled	0.0747088730315122	201.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo_novus	promo_controlled	6.717085740033105e-08	48.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo_novus	promo_controlled	0.7263837848406884	56.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
milk	silpo_novus	promo_controlled	daily	none	none	n/a	n/a	1803	nan	nan	nan	nan	nan	nan	nan	unavailable	1	No admissible model estim...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
none	none	n/a	n/a	nan	nan	nan	1

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	4.889813923695362	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-02	4.880939529121857	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-03	4.873865236176802	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-04	4.868432562653949	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-05	4.864483026347049	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-06	4.861858145049856	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-07	4.860399436556123	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-08	4.8599484186596005	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-09	4.861523687586162	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-10	4.865852176359705	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-11	4.872337907971017	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-12	4.880384905410883	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-13	4.889397191670094	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-14	4.898778789739431	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-15	4.907933722609686	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-16	4.919261740789084	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-17	4.93424761149779	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-18	4.950922882794503	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-19	4.967319102737919	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-20	4.981467819386735	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-21	4.991400580799647	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-22	4.995148935035351	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-23	4.937254183107472	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-24	4.7879466649776745	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-25	4.583791487126728	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-26	4.361353756035397	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-27	4.157198578184451	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-28	4.0078910600546545	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-01	3.949996308126774	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-02	3.9561770869121475	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-03	3.972116990095481	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-04	3.9939123679175887	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-05	4.017659570619289	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-06	4.039454948441397	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-07	4.05539485162473	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-08	4.061575630410104	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-09	4.058274307819267	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-10	4.049435990260466	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-11	4.036659153054266	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-12	4.021542271521231	nan	nan	milk	silpo_novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	-0.1619954987840599	
producer_to_prozorro	2	-0.1596960610741088	
producer_to_prozorro	3	-0.1550755940895887	
producer_to_prozorro	4	-0.14782606735994822	
producer_to_prozorro	5	-0.13755773452175912	
producer_to_prozorro	6	-0.12184188698467123	
producer_to_prozorro	7	-0.1052691917763927	
producer_to_prozorro	8	-0.08910616067044506	
producer_to_prozorro	9	-0.07590716773933302	
producer_to_prozorro	10	-0.06623846374994181	
producer_to_prozorro	11	-0.0604388966635031	
producer_to_prozorro	12	-0.053445772573508486	
producer_to_prozorro	13	-0.044442939499120704	
producer_to_prozorro	14	-0.03488503971137872	
producer_to_prozorro	15	-0.027271625330034636	
producer_to_prozorro	16	-0.02250000775439031	
producer_to_prozorro	17	-0.021139446186402542	
producer_to_prozorro	18	-0.017275889530894442	
producer_to_prozorro	19	-0.0052249478693689524	
producer_to_prozorro	20	0.012485029223047036	
producer_to_prozorro	21	0.03347787604499084	
producer_to_prozorro	22	0.05374998643553993	
producer_to_prozorro	23	0.06948873713684375	
producer_to_prozorro	24	0.07823993038535337	
producer_to_prozorro	25	0.08172443030355239	
producer_to_prozorro	26	0.08396988659951972	
producer_to_prozorro	27	0.0844840372389968	
producer_to_prozorro	28	0.08380704389180071	
producer_to_prozorro	29	0.08186704327105128	
producer_to_prozorro	30	0.07802717791370897	
prozorro_to_retail	1	0.058986821056920116	
prozorro_to_retail	2	0.06596364838209384	
prozorro_to_retail	3	0.06614951430425206	
prozorro_to_retail	4	-0.021768827279205462	
prozorro_to_retail	5	0.001715675194402283	
prozorro_to_retail	6	0.02920914546748517	
prozorro_to_retail	7	0.02810615541100227	
prozorro_to_retail	8	-0.04145539795472525	
prozorro_to_retail	9	0.1628834716667106	
prozorro_to_retail	10	-0.13450767385630638	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
milk	silpo_novus	promo_controlled	daily	I(1)	I(1)	I(2)	1793	211	59	201	48	1	Rejected for ARDL/ECM due...

Table: NARDL_Multipliers

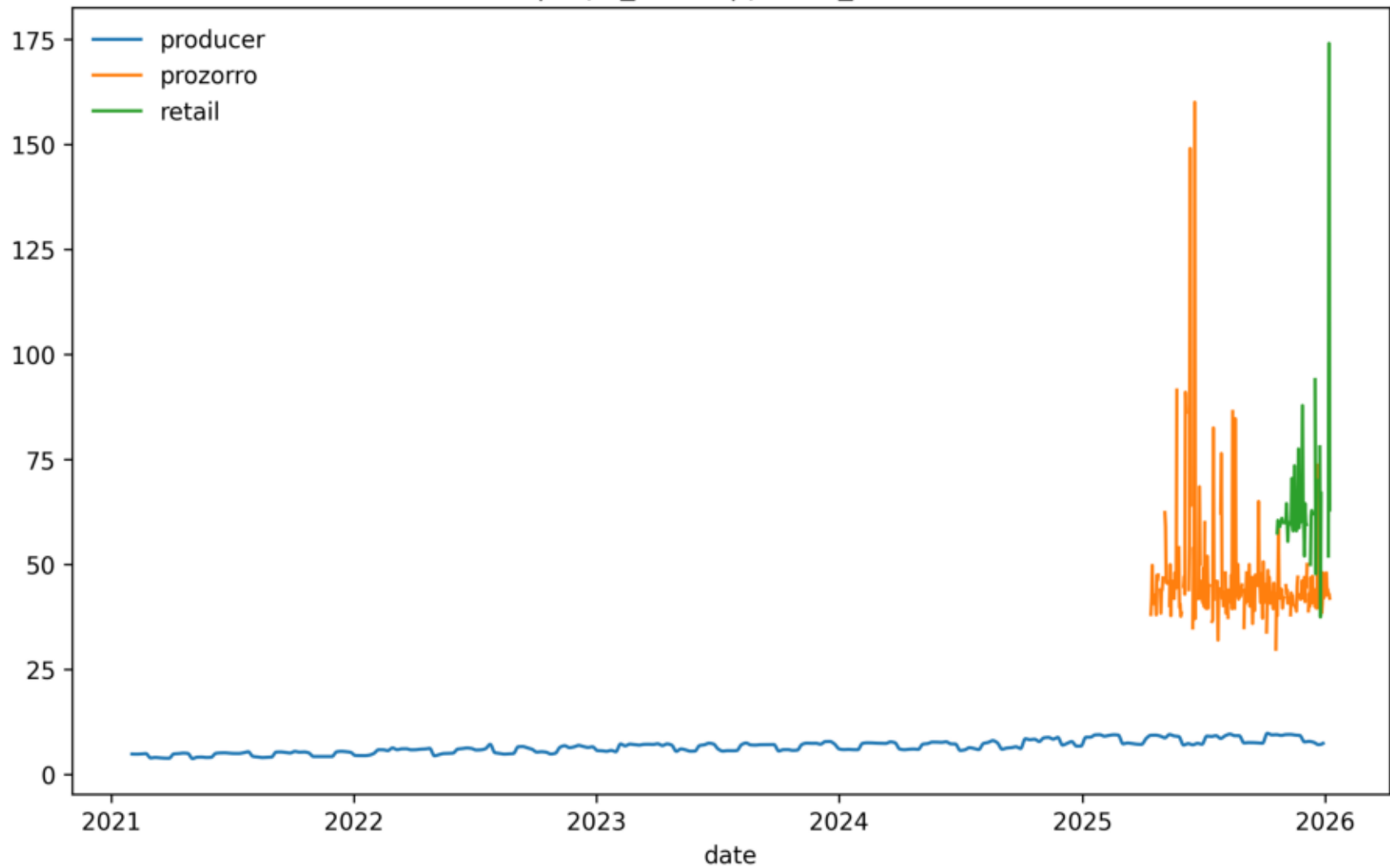
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Table: VECM_IRF

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time_series_promo_controlled.png

milk | silpo_novus | promo_controlled



Lag-Correlation Profile | milk | silpo_novus | promo_controlled

